

Implementation Approach for Digital Support System (SAC)

1. Project Initiation & Planning

- Kick-off workshops with Department & stakeholders
- Finalize FRS & SRS
- Define acceptance criteria
- Set up governance, communication plan, risk register
- Prepare detailed project plan with milestones

2. System Design

- 3-tier architecture (Frontend – API – Database)
- Frontend: React.js (bilingual, responsive)
- Backend: Node.js (Express/Nest, REST APIs)
- Database: PostgreSQL/MySQL
- Security: Role-based access, 2FA, SSL, CERT-In audit
- Integrations: Google Maps, Biometric/GPS, SMS/Email/WhatsApp

3. Development & Module-Wise Implementation

Agile Iterative Development – deliver in sprints (2–3 weeks each), with demo to Department after each sprint.

Phase 1 – Foundation:

- User Management (registration, login, OTP, roles)
- Knowledge Management
- Audit & Compliance framework

Phase 2 – Core Student Services:

- Student Management
- Grievance Redressal (SLA, escalation)
- Communication & Notifications
- Feedback Monitoring

Phase 3 – Operations & Analytics:

- Geo Mapping
- Attendance Management
- MIS & Reporting
- Admin Dashboards

Phase 4 – AI & Helpdesk:

- AI-powered WhatsApp Chatbot

- Support & Helpdesk
- Feedback Monitoring

4. Testing & Quality Assurance

- Unit testing
- Integration testing
- Performance testing
- Security testing (CERT-In audit)
- UAT with Department

5. Deployment

- Staging deployment for UAT
- Production deployment on GCC cloud
- Domain setup, SSL, DNS
- Go-Live after approval

6. Operations & Maintenance (O&M)

- Helpdesk Support (L1/L2)
- Regular updates, patches
- Performance monitoring
- Compliance audits
- Change Requests (CRs) handling
- Monthly/Quarterly O&M reports

7. Implementation Timeline (High-Level)

Phase	Activities
Phase 1	Requirement Gathering, SRS, System Design
Phase 2	Core Development (User Management, Student Management, Grievances)
Phase 3	Operations Modules (Geo Mapping, Attendance, MIS, Dashboards)
Phase 4	AI Chatbot, Helpdesk, Feedback Monitoring
Phase 5	Testing, Security Audit, UAT
Phase 6	Deployment & Go-Live
Phase 7	O&M Support (24 months)

List of 3rd party APIs:

- Google Maps API
- WhatsApp Business API
- SMS/Email Gateway APIs
- Biometric SDK / GPS APIs
- NLP/Chatbot APIs
- Cloud Hosting APIs

To ensure robust security of the Digital Support System, both the frontend and backend will incorporate multiple layers of protection. On the frontend, all communication with the server will be enforced over HTTPS with TLS 1.3, user sessions will be secured with JWT tokens stored in Http only cookies, and sensitive user inputs will be validated and sanitized to prevent XSS or injection attacks. Role-based access will control visibility of modules and data based on user type (student, facilitator, nodal officer, administrator). On the backend, sensitive data such as student records, grievance logs, and attendance details will be encrypted at rest using AES-256, while authentication credentials will be hashed using bcrypt/argon2. The APIs will be protected with strict authentication, rate limiting, and input validation to guard against SQL injection, CSRF, and brute force attempts. Comprehensive audit logs will be maintained to track every transaction, and periodic CERT-In audits along with compliance to the DPDP Act will ensure ongoing security and data privacy.

Data Security Management

1. Identity & Access Management (IAM)

- Use Principle of Least Privilege (PoLP) – give developers only the permissions they need.
- Enable IAM Recommender in GCP to remove unused permissions.

2. Secrets Management

- Store secrets in Secret Manager
- Never hardcode secrets in Bitbucket / CI/CD pipelines.

3. Networking Security

- Use VPC Service Controls to protect against data exfiltration.
- Enable Firewall Rules with least privilege.

4. Storage Security

- Use Bucket Lock and Object Versioning in Cloud Storage to prevent accidental deletion.
- Enable Data Loss Prevention (DLP API) to scan sensitive data.

5. Monitoring

- Configured Cloud Monitoring & Logging for performance tracking and alerting.
- Enable Cloud Audit Logs for all admin & data access activities.

6. CI/CD Security

- Store Docker images in Artifact Registry with vulnerability scanning enabled.
- Use Cloud Build with IAM-controlled service accounts (not default accounts).

LLM Models for Chatbot

- Lamma 3
- Open AI

Cloud Server Pricing

Cloud run - 37.92\$ per month